

19 Touch sensing controller (TSC)

19.1 Introduction

The touch sensing controller provides a simple solution for adding capacitive sensing functionality to any application. Capacitive sensing technology is able to detect finger presence near an electrode that is protected from direct touch by a dielectric (for example glass, plastic). The capacitive variation introduced by the finger (or any conductive object) is measured using a proven implementation based on a surface charge transfer acquisition principle.

The touch sensing controller is fully supported by the STMTouch touch sensing firmware library, which is free to use and allows touch sensing functionality to be implemented reliably in the end application.

19.2 TSC main features

The touch sensing controller has the following main features:

- Proven and robust surface charge transfer acquisition principle
- Supports up to 24 capacitive sensing channels
- Up to 8 capacitive sensing channels can be acquired in parallel offering a very good response time
- Spread spectrum feature to improve system robustness in noisy environments
- Full hardware management of the charge transfer acquisition sequence
- Programmable charge transfer frequency
- Programmable sampling capacitor I/O pin
- Programmable channel I/O pin
- Programmable max count value to avoid long acquisition when a channel is faulty
- Dedicated end of acquisition and max count error flags with interrupt capability
- One sampling capacitor for up to 3 capacitive sensing channels to reduce the system components
- Compatible with proximity, touchkey, linear and rotary touch sensor implementation
- Designed to operate with STMTouch touch sensing firmware library

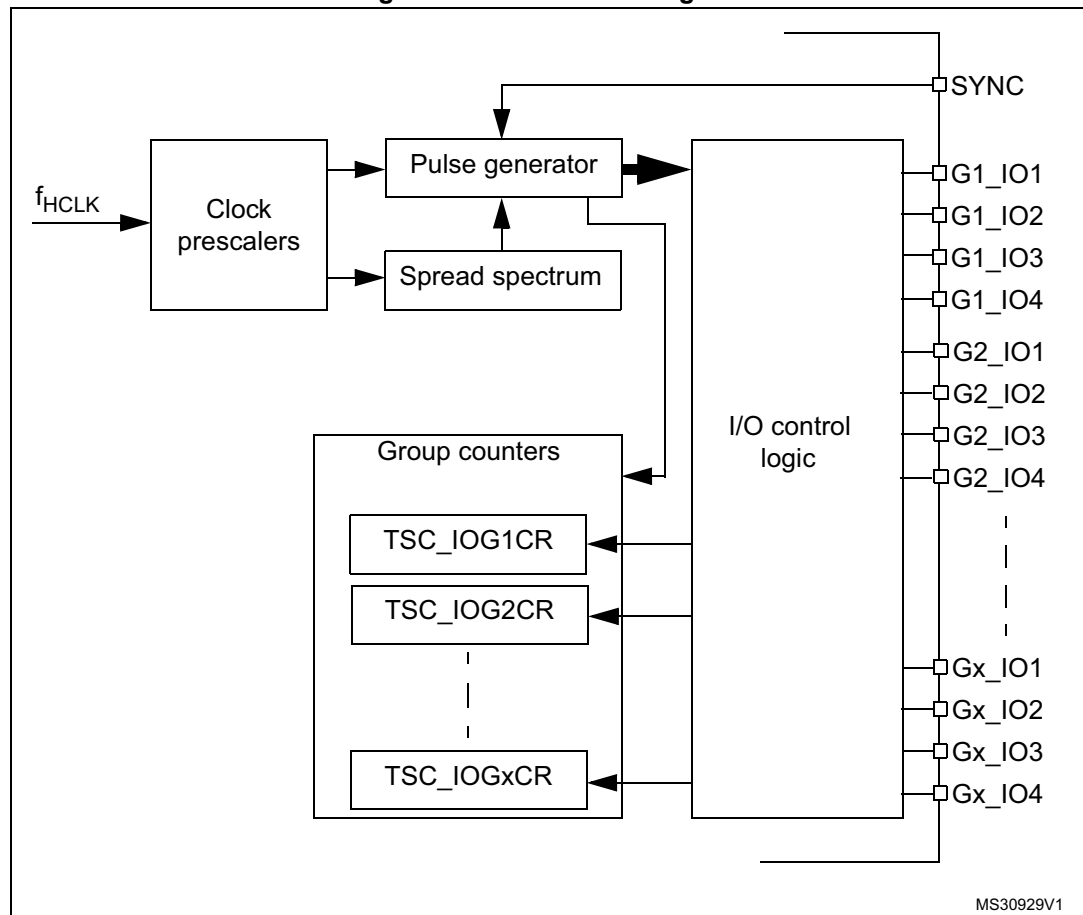
Note: The number of capacitive sensing channels is dependent on the size of the packages and subject to IO availability.

19.3 TSC functional description

19.3.1 TSC block diagram

The block diagram of the touch sensing controller is shown in [Figure 135](#).

Figure 135. TSC block diagram



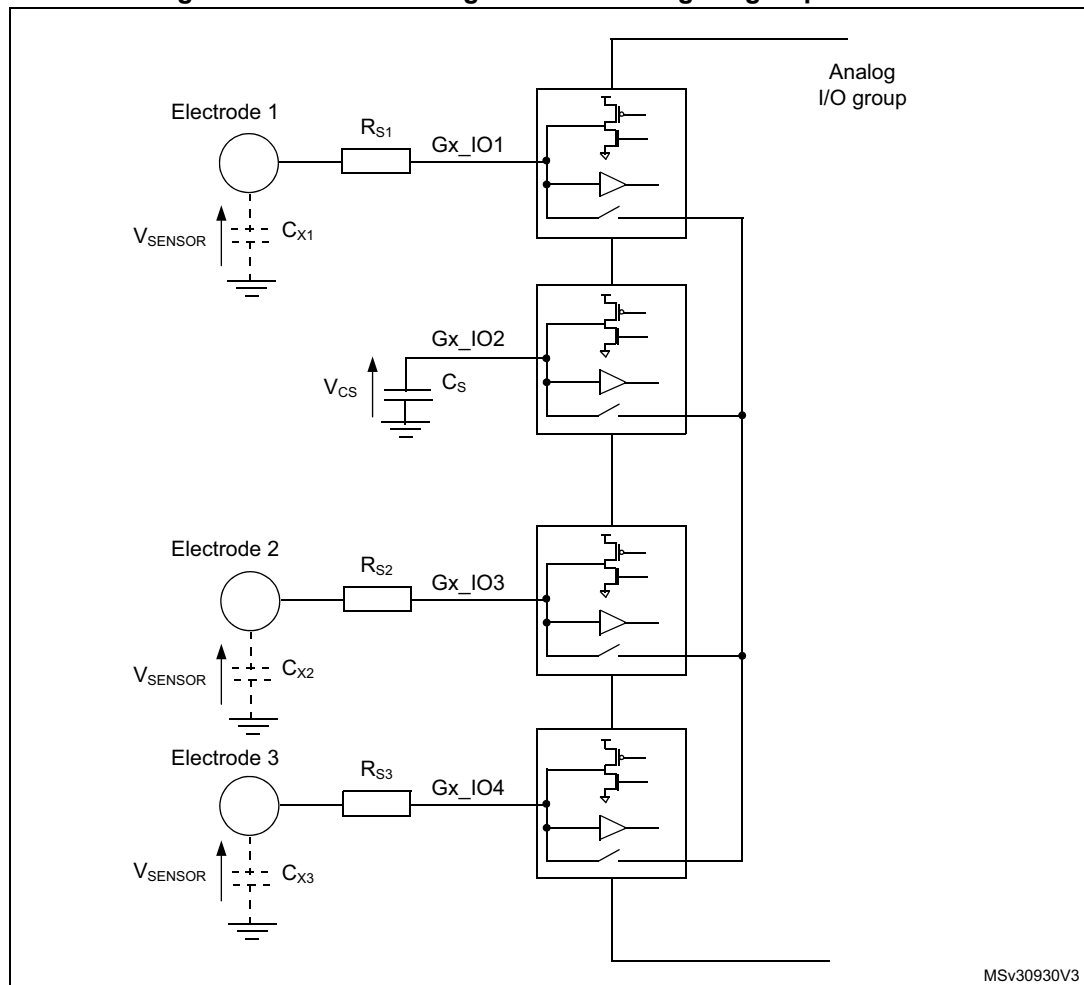
19.3.2 Surface charge transfer acquisition overview

The surface charge transfer acquisition is a proven, robust and efficient way to measure a capacitance. It uses a minimum number of external components to operate with a single ended electrode type. This acquisition is designed around an analog I/O group composed of up to four GPIOs (see [Figure 136](#)). Several analog I/O groups are available to allow the acquisition of several capacitive sensing channels simultaneously and to support a larger number of capacitive sensing channels. Within a same analog I/O group, the acquisition of the capacitive sensing channels is sequential.

One of the GPIOs is dedicated to the sampling capacitor C_S . Only one sampling capacitor I/O per analog I/O group must be enabled at a time.

The remaining GPIOs are dedicated to the electrodes and are commonly called channels. For some specific needs (such as proximity detection), it is possible to simultaneously enable more than one channel per analog I/O group.

Figure 136. Surface charge transfer analog I/O group structure



MSv30930V3

Note: Gx_IOy where x is the analog I/O group number and y the GPIO number within the selected group.

The surface charge transfer acquisition principle consists of charging an electrode capacitance (C_X) and transferring a part of the accumulated charge into a sampling capacitor (C_S). This sequence is repeated until the voltage across C_S reaches a given threshold (V_{IH} in our case). The number of charge transfers required to reach the threshold is a direct representation of the size of the electrode capacitance.

[Table 113](#) details the charge transfer acquisition sequence of the capacitive sensing channel 1. States 3 to 7 are repeated until the voltage across C_S reaches the given threshold. The same sequence applies to the acquisition of the other channels. The electrode serial resistor R_S improves the ESD immunity of the solution.

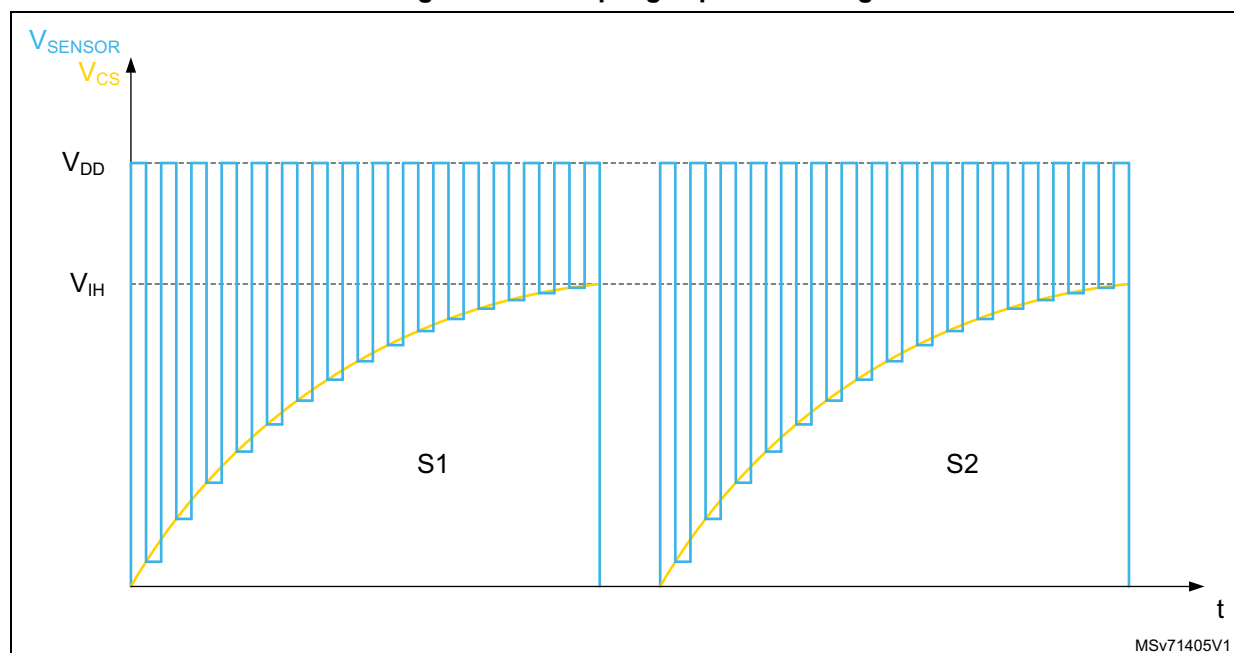
Table 113. Acquisition sequence summary

State	Gx_IO1 (channel)	Gx_IO2 (sampling)	Gx_IO3 (channel)	Gx_IO4 (channel)	State description
#1	Input floating with analog switch closed	Output open- drain low with analog switch closed	Input floating with analog switch closed		Discharge all C _X and C _S
#2	Input floating				Dead time
#3	Output push- pull high	Input floating			Charge C _{X1}
#4	Input floating				Dead time
#5	Input floating with analog switch closed		Input floating		Charge transfer from C _{X1} to C _S
#6	Input floating				Dead time
#7	Input floating				Measure C _S voltage

Note: Gx_IOy where x is the analog I/O group number and y the GPIO number within the selected group.

The voltage variation over the time on the sampling capacitor C_S is detailed below (refer to [Figure 136](#) for V_{SENSOR} and V_{CS} definition):

Figure 137. Sampling capacitor voltage variation



19.3.3 Reset and clocks

The TSC clock source is the AHB clock (HCLK). Two programmable prescalers are used to generate the pulse generator and the spread spectrum internal clocks:

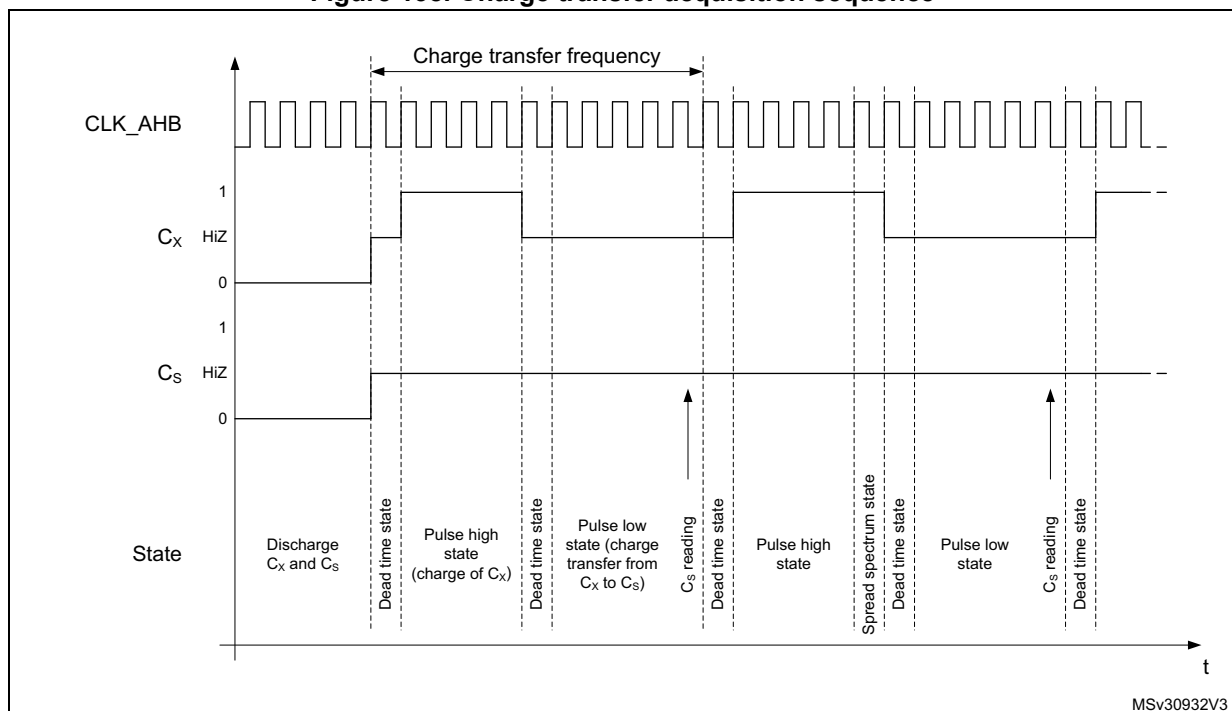
- The pulse generator clock (PGCLK) is defined using the PGPSC[2:0] bits of the TSC_CR register
- The spread spectrum clock (SSCLK) is defined using the SSPSC bit of the TSC_CR register

The reset and clock controller (RCC) provides dedicated bits to enable the touch sensing controller clock and to reset this peripheral. For more information, refer to [Section 9: Reset and clock control \(RCC\)](#).

19.3.4 Charge transfer acquisition sequence

An example of a charge transfer acquisition sequence is detailed in [Figure 138](#).

Figure 138. Charge transfer acquisition sequence



For higher flexibility, the charge transfer frequency is fully configurable. Both the pulse high state (charge of C_x) and the pulse low state (transfer of charge from C_x to C_s) duration can be defined using the CTPH[3:0] and CTPL[3:0] bits in the TSC_CR register. The standard range for the pulse high and low states duration is 500 ns to 2 μs. To ensure a correct measurement of the electrode capacitance, the pulse high state duration must be set to ensure that C_x is always fully charged.

A dead time where both the sampling capacitor I/O and the channel I/O are in input floating state is inserted between the pulse high and low states to ensure an optimum charge transfer acquisition sequence. This state duration is 2 periods of HCLK.

At the end of the pulse high state and if the spread spectrum feature is enabled, a variable number of periods of the SSCLK clock are added.

The reading of the sampling capacitor I/O, to determine if the voltage across C_S has reached the given threshold, is performed at the end of the pulse low state.

Note: The following TSC control register configurations are forbidden:

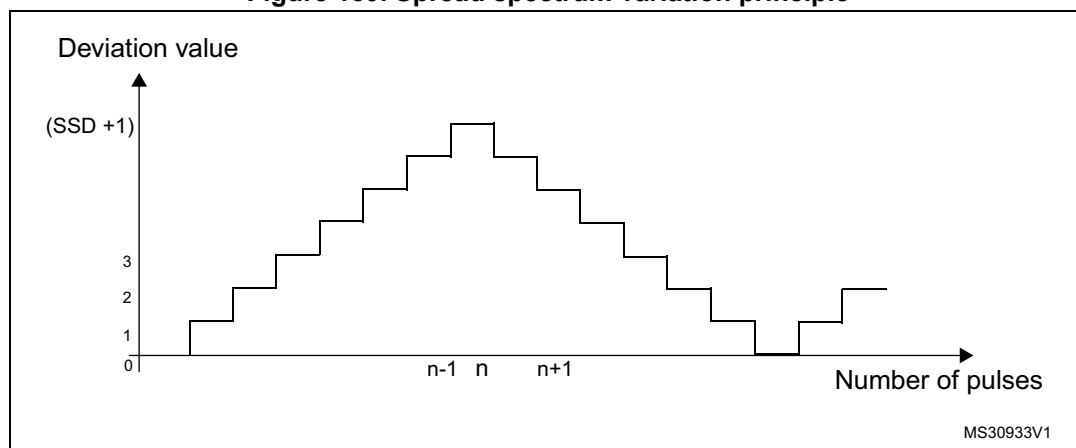
- *bits PGPSC are set to '000' and bits CTPL are set to '0000'*
- *bits PGPSC are set to '000' and bits CTPL are set to '0001'*
- *bits PGPSC are set to '001' and bits CTPL are set to '0000'*

19.3.5 Spread spectrum feature

The spread spectrum feature generates a variation of the charge transfer frequency. This is done to improve the robustness of the charge transfer acquisition in noisy environments and also to reduce the induced emission. The maximum frequency variation is in the range of 10% to 50% of the nominal charge transfer period. For instance, for a nominal charge transfer frequency of 250 kHz (4 μ s), the typical spread spectrum deviation is 10% (400 ns) which leads to a minimum charge transfer frequency of ~227 kHz.

In practice, the spread spectrum consists of adding a variable number of SSCLK periods to the pulse high state using the principle shown below:

Figure 139. Spread spectrum variation principle



The table below details the maximum frequency deviation with different HCLK settings:

Table 114. Spread spectrum deviation versus AHB clock frequency

f_{HCLK}	Spread spectrum step	Maximum spread spectrum deviation
24 MHz	41.6 ns	10666.6 ns
48 MHz	20.8 ns	5333.3 ns

The spread spectrum feature can be disabled/enabled using the SSE bit in the TSC_CR register. The frequency deviation is also configurable to accommodate the device HCLK clock frequency and the selected charge transfer frequency through the SSPSC and SSD[6:0] bits in the TSC_CR register.

19.3.6 Max count error

The max count error prevents long acquisition times resulting from a faulty capacitive sensing channel. It consists of specifying a maximum count value for the analog I/O group counters. This maximum count value is specified using the MCV[2:0] bits in the TSC_CR register. As soon as an acquisition group counter reaches this maximum value, the ongoing acquisition is stopped and the end of acquisition (EOAF bit) and max count error (MCEF bit) flags are both set. An interrupt can also be generated if the corresponding end of acquisition (EOAIE bit) or/and max count error (MCEIE bit) interrupt enable bits are set.

19.3.7 Sampling capacitor I/O and channel I/O mode selection

To allow the GPIOs to be controlled by the touch sensing controller, the corresponding alternate function must be enabled through the standard GPIO registers and the GPIOxAFR registers.

The GPIOs modes controlled by the TSC are defined using the TSC_IOSCR and TSC_IOCCR register.

When there is no ongoing acquisition, all the I/Os controlled by the touch sensing controller are in default state. While an acquisition is ongoing, only unused I/Os (neither defined as sampling capacitor I/O nor as channel I/O) are in default state. The IODEF bit in the TSC_CR register defines the configuration of the I/Os which are in default state. The table below summarizes the configuration of the I/O depending on its mode.

Table 115. I/O state depending on its mode and IODEF bit value

IODEF bit	Acquisition status	Unused I/O mode	Channel I/O mode	Sampling capacitor I/O mode
0 (output push-pull low)	No	Output push-pull low	Output push-pull low	Output push-pull low
0 (output push-pull low)	Ongoing	Output push-pull low	-	-
1 (input floating)	No	Input floating	Input floating	Input floating
1 (input floating)	Ongoing	Input floating	-	-

Unused I/O mode

An unused I/O corresponds to a GPIO controlled by the TSC peripheral but not defined as an electrode I/O nor as a sampling capacitor I/O.

Sampling capacitor I/O mode

To allow the control of the sampling capacitor I/O by the TSC peripheral, the corresponding GPIO must be first set to alternate output open drain mode and then the corresponding Gx_IOy bit in the TSC_IOSCR register must be set.

Only one sampling capacitor per analog I/O group must be enabled at a time.

Channel I/O mode

To allow the control of the channel I/O by the TSC peripheral, the corresponding GPIO must be first set to alternate output push-pull mode and the corresponding Gx_IOy bit in the TSC_IOCCR register must be set.

For proximity detection where a higher equivalent electrode surface is required or to speed-up the acquisition process, it is possible to enable and simultaneously acquire several channels belonging to the same analog I/O group.

Note: *During the acquisition phase and even if the TSC peripheral alternate function is not enabled, as soon as the TSC_IOSCR or TSC_IOCCR bit is set, the corresponding GPIO analog switch is automatically controlled by the touch sensing controller.*

19.3.8 Acquisition mode

The touch sensing controller offers two acquisition modes:

- Normal acquisition mode: the acquisition starts as soon as the START bit in the TSC_CR register is set.
- Synchronized acquisition mode: the acquisition is enabled by setting the START bit in the TSC_CR register but only starts upon the detection of a falling edge or a rising edge and high level on the SYNC input pin. This mode is useful for synchronizing the capacitive sensing channels acquisition with an external signal without additional CPU load.

The GxE bits in the TSC_I OGCSR registers specify which analog I/O groups are enabled (corresponding counter is counting). The C_S voltage of a disabled analog I/O group is not monitored and this group does not participate in the triggering of the end of acquisition flag. However, if the disabled analog I/O group contains some channels, they are pulsed.

When the C_S voltage of an enabled analog I/O group reaches the given threshold, the corresponding GxS bit of the TSC_I OGCSR register is set. When the acquisition of all enabled analog I/O groups is complete (all GxS bits of all enabled analog I/O groups are set), the EOAF flag in the TSC_ISR register is set. An interrupt request is generated if the EOAI E bit in the TSC_I ER register is set.

In the case that a max count error is detected, the ongoing acquisition is stopped and both the EOAF and MCEF flags in the TSC_ISR register are set. Interrupt requests can be generated for both events if the corresponding bits (EOAI E and MCEI E bits of the TSCI ER register) are set. Note that when the max count error is detected the remaining GxS bits in the enabled analog I/O groups are not set.

To clear the interrupt flags, the corresponding EOAI C and MCEI C bits in the TSC_I CR register must be set.

The analog I/O group counters are cleared when a new acquisition is started. They are updated with the number of charge transfer cycles generated on the corresponding channel(s) upon the completion of the acquisition.

19.3.9 I/O hysteresis and analog switch control

In order to offer a higher flexibility, the touch sensing controller is able to take the control of the Schmitt trigger hysteresis and analog switch of each Gx_I O_y. This control is available whatever the I/O control mode is (controlled by standard GPIO registers or other peripherals) assuming that the touch sensing controller is enabled. This may be useful to perform a different acquisition sequence or for other purposes.

In order to improve the system immunity, the Schmitt trigger hysteresis of the GPIOs controlled by the TSC must be disabled by resetting the corresponding Gx_I O_y bit in the TSC_I OHCR register.

19.4 TSC low-power modes

Table 116. Effect of low-power modes on TSC

Mode	Description
Sleep	No effect TSC interrupts cause the device to exit Sleep mode.
Stop	TSC registers are frozen
Standby	The TSC stops its operation until the Stop or Standby mode is exited.

19.5 TSC interrupts

Table 117. Interrupt control bits

Interrupt event	Enable control bit	Event flag	Clear flag bit	Exit the Sleep mode	Exit the Stop mode	Exit the Standby mode
End of acquisition	EOAIE	EOAIF	EOAIC	Yes	No	No
Max count error	MCEIE	MCEIF	MCEIC	Yes	No	No

19.6 TSC registers

Refer to [Section 2.2 on page 47](#) of the reference manual for a list of abbreviations used in register descriptions.

The peripheral registers can be accessed by words (32-bit).

19.6.1 TSC control register (TSC_CR)

Address offset: 0x00

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CTPH[3:0]				CTPL[3:0]				SSD[6:0]						SSE	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SSPSC	PGPSC[2:0]			Res.	Res.	Res.	Res.	MCV[2:0]			IODEF	SYNC POL	AM	START	TSCE
rw	rw	rw	rw					rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:28 **CTPH[3:0]**: Charge transfer pulse high

These bits are set and cleared by software. They define the duration of the high state of the charge transfer pulse (charge of C_X).

0000: $1 \times t_{PGCLK}$

0001: $2 \times t_{PGCLK}$

...

1111: $16 \times t_{PGCLK}$

Note: These bits must not be modified when an acquisition is ongoing.

Bits 27:24 **CTPL[3:0]**: Charge transfer pulse low

These bits are set and cleared by software. They define the duration of the low state of the charge transfer pulse (transfer of charge from C_X to C_S).

0000: $1 \times t_{PGCLK}$

0001: $2 \times t_{PGCLK}$

...

1111: $16 \times t_{PGCLK}$

Note: These bits must not be modified when an acquisition is ongoing.

Note: Some configurations are forbidden. Refer to the [Section 19.3.4: Charge transfer acquisition sequence](#) for details.

Bits 23:17 **SSD[6:0]**: Spread spectrum deviation

These bits are set and cleared by software. They define the spread spectrum deviation which consists in adding a variable number of periods of the SSCLK clock to the charge transfer pulse high state.

0000000: $1 \times t_{SSCLK}$

0000001: $2 \times t_{SSCLK}$

...

1111111: $128 \times t_{SSCLK}$

Note: These bits must not be modified when an acquisition is ongoing.

Bit 16 **SSE**: Spread spectrum enable

This bit is set and cleared by software to enable/disable the spread spectrum feature.

0: Spread spectrum disabled

1: Spread spectrum enabled

Note: This bit must not be modified when an acquisition is ongoing.

Bit 15 **SSPSC**: Spread spectrum prescaler

This bit is set and cleared by software. It selects the AHB clock divider used to generate the spread spectrum clock (SSCLK).

0: f_{HCLK}

1: $f_{HCLK} / 2$

Note: This bit must not be modified when an acquisition is ongoing.

Bits 14:12 **PGPSC[2:0]**: Pulse generator prescaler

These bits are set and cleared by software. They select the AHB clock divider used to generate the pulse generator clock (PGCLK).

000: f_{HCLK}
 001: $f_{HCLK} / 2$
 010: $f_{HCLK} / 4$
 011: $f_{HCLK} / 8$
 100: $f_{HCLK} / 16$
 101: $f_{HCLK} / 32$
 110: $f_{HCLK} / 64$
 111: $f_{HCLK} / 128$

Note: These bits must not be modified when an acquisition is ongoing.

Note: Some configurations are forbidden. Refer to the [Section 19.3.4: Charge transfer acquisition sequence](#) for details.

Bits 11:8 Reserved, must be kept at reset value.

Bits 7:5 **MCV[2:0]**: Max count value

These bits are set and cleared by software. They define the maximum number of charge transfer pulses that can be generated before a max count error is generated.

000: 255
 001: 511
 010: 1023
 011: 2047
 100: 4095
 101: 8191
 110: 16383
 111: reserved

Note: These bits must not be modified when an acquisition is ongoing.

Bit 4 **IODEF**: I/O Default mode

This bit is set and cleared by software. It defines the configuration of all the TSC I/Os when there is no ongoing acquisition. When there is an ongoing acquisition, it defines the configuration of all unused I/Os (not defined as sampling capacitor I/O or as channel I/O).

0: I/Os are forced to output push-pull low
 1: I/Os are in input floating

Note: This bit must not be modified when an acquisition is ongoing.

Bit 3 **SYNCPOL**: Synchronization pin polarity

This bit is set and cleared by software to select the polarity of the synchronization input pin.

0: Falling edge only
 1: Rising edge and high level

Bit 2 **AM**: Acquisition mode

This bit is set and cleared by software to select the acquisition mode.

0: Normal acquisition mode (acquisition starts as soon as START bit is set)

1: Synchronized acquisition mode (acquisition starts if START bit is set and when the selected signal is detected on the SYNC input pin)

Note: This bit must not be modified when an acquisition is ongoing.

Bit 1 **START**: Start a new acquisition

This bit is set by software to start a new acquisition. It is cleared by hardware as soon as the acquisition is complete or by software to cancel the ongoing acquisition.

0: Acquisition not started

1: Start a new acquisition

Bit 0 **TSCE**: Touch sensing controller enable

This bit is set and cleared by software to enable/disable the touch sensing controller.

0: Touch sensing controller disabled

1: Touch sensing controller enabled

Note: When the touch sensing controller is disabled, TSC registers settings have no effect.

19.6.2 TSC interrupt enable register (TSC_IER)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCEIE	EOAIE
														rw	rw

Bits 31:2 Reserved, must be kept at reset value.

Bit 1 **MCEIE**: Max count error interrupt enable

This bit is set and cleared by software to enable/disable the max count error interrupt.

0: Max count error interrupt disabled

1: Max count error interrupt enabled

Bit 0 **EOAIE**: End of acquisition interrupt enable

This bit is set and cleared by software to enable/disable the end of acquisition interrupt.

0: End of acquisition interrupt disabled

1: End of acquisition interrupt enabled

19.6.3 TSC interrupt clear register (TSC_ICR)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCEIC	EOAIC
														rw	rw

Bits 31:2 Reserved, must be kept at reset value.

Bit 1 **MCEIC**: Max count error interrupt clear

This bit is set by software to clear the max count error flag and it is cleared by hardware when the flag is reset. Writing a '0' has no effect.

0: No effect

1: Clears the corresponding MCEF of the TSC_ISR register

Bit 0 **EOAIC**: End of acquisition interrupt clear

This bit is set by software to clear the end of acquisition flag and it is cleared by hardware when the flag is reset. Writing a '0' has no effect.

0: No effect

1: Clears the corresponding EOAF of the TSC_ISR register

19.6.4 TSC interrupt status register (TSC_ISR)

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCEF	EOAF
														r	r

Bits 31:2 Reserved, must be kept at reset value.

Bit 1 **MCEF**: Max count error flag

This bit is set by hardware as soon as an analog I/O group counter reaches the max count value specified. It is cleared by software writing 1 to the bit MCEIC of the TSC_ICR register.

0: No max count error (MCE) detected

1: Max count error (MCE) detected

Bit 0 **EOAF**: End of acquisition flag

This bit is set by hardware when the acquisition of all enabled group is complete (all GxS bits of all enabled analog I/O groups are set or when a max count error is detected). It is cleared by software writing 1 to the bit EOAIC of the TSC_ICR register.

0: Acquisition is ongoing or not started

1: Acquisition is complete

19.6.5 TSC I/O hysteresis control register (TSC_IOHCR)

Address offset: 0x10

Reset value: 0xFFFF FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
G8_IO4	G8_IO3	G8_IO2	G8_IO1	G7_IO4	G7_IO3	G7_IO2	G7_IO1	G6_IO4	G6_IO3	G6_IO2	G6_IO1	G5_IO4	G5_IO3	G5_IO2	G5_IO1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
G4_IO4	G4_IO3	G4_IO2	G4_IO1	G3_IO4	G3_IO3	G3_IO2	G3_IO1	G2_IO4	G2_IO3	G2_IO2	G2_IO1	G1_IO4	G1_IO3	G1_IO2	G1_IO1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **Gx_IOy**: Gx_IOy Schmitt trigger hysteresis mode

These bits are set and cleared by software to enable/disable the Gx_IOy Schmitt trigger hysteresis.

0: Gx_IOy Schmitt trigger hysteresis disabled

1: Gx_IOy Schmitt trigger hysteresis enabled

Note: These bits control the I/O Schmitt trigger hysteresis whatever the I/O control mode is (even if controlled by standard GPIO registers).

19.6.6 TSC I/O analog switch control register (TSC_IOASCR)

Address offset: 0x18

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
G8_IO4	G8_IO3	G8_IO2	G8_IO1	G7_IO4	G7_IO3	G7_IO2	G7_IO1	G6_IO4	G6_IO3	G6_IO2	G6_IO1	G5_IO4	G5_IO3	G5_IO2	G5_IO1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
G4_IO4	G4_IO3	G4_IO2	G4_IO1	G3_IO4	G3_IO3	G3_IO2	G3_IO1	G2_IO4	G2_IO3	G2_IO2	G2_IO1	G1_IO4	G1_IO3	G1_IO2	G1_IO1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **Gx_IOy**: Gx_IOy analog switch enable

These bits are set and cleared by software to enable/disable the Gx_IOy analog switch.

0: Gx_IOy analog switch disabled (opened)

1: Gx_IOy analog switch enabled (closed)

Note: These bits control the I/O analog switch whatever the I/O control mode is (even if controlled by standard GPIO registers).

19.6.7 TSC I/O sampling control register (TSC_IOSCR)

Address offset: 0x20

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
G8_IO4	G8_IO3	G8_IO2	G8_IO1	G7_IO4	G7_IO3	G7_IO2	G7_IO1	G6_IO4	G6_IO3	G6_IO2	G6_IO1	G5_IO4	G5_IO3	G5_IO2	G5_IO1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
G4_IO4	G4_IO3	G4_IO2	G4_IO1	G3_IO4	G3_IO3	G3_IO2	G3_IO1	G2_IO4	G2_IO3	G2_IO2	G2_IO1	G1_IO4	G1_IO3	G1_IO2	G1_IO1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **Gx_IOy**: Gx_IOy sampling mode

These bits are set and cleared by software to configure the Gx_IOy as a sampling capacitor I/O. Only one I/O per analog I/O group must be defined as sampling capacitor.

0: Gx_IOy unused

1: Gx_IOy used as sampling capacitor

Note: These bits must not be modified when an acquisition is ongoing.

During the acquisition phase and even if the TSC peripheral alternate function is not enabled, as soon as the TSC_IOSCR bit is set, the corresponding GPIO analog switch is automatically controlled by the touch sensing controller.

19.6.8 TSC I/O channel control register (TSC_IOCOCR)

Address offset: 0x28

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
G8_IO4	G8_IO3	G8_IO2	G8_IO1	G7_IO4	G7_IO3	G7_IO2	G7_IO1	G6_IO4	G6_IO3	G6_IO2	G6_IO1	G5_IO4	G5_IO3	G5_IO2	G5_IO1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
G4_IO4	G4_IO3	G4_IO2	G4_IO1	G3_IO4	G3_IO3	G3_IO2	G3_IO1	G2_IO4	G2_IO3	G2_IO2	G2_IO1	G1_IO4	G1_IO3	G1_IO2	G1_IO1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **Gx_IOy**: Gx_IOy channel mode

These bits are set and cleared by software to configure the Gx_IOy as a channel I/O.

0: Gx_IOy unused

1: Gx_IOy used as channel

Note: These bits must not be modified when an acquisition is ongoing.

During the acquisition phase and even if the TSC peripheral alternate function is not enabled, as soon as the TSC_IOCOCR bit is set, the corresponding GPIO analog switch is automatically controlled by the touch sensing controller.

19.6.9 TSC I/O group control status register (TSC_I OGCSR)

Address offset: 0x30

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	G8S	G7S	G6S	G5S	G4S	G3S	G2S	G1S
								r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	G8E	G7E	G6E	G5E	G4E	G3E	G2E	G1E
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:24 Reserved, must be kept at reset value.

Bits 23:16 **GxS**: Analog I/O group x status

These bits are set by hardware when the acquisition on the corresponding enabled analog I/O group x is complete. They are cleared by hardware when a new acquisition is started.

0: Acquisition on analog I/O group x is ongoing or not started

1: Acquisition on analog I/O group x is complete

Note: When a max count error is detected the remaining GxS bits of the enabled analog I/O groups are not set.

Bits 15:8 Reserved, must be kept at reset value.

Bits 7:0 **GxE**: Analog I/O group x enable

These bits are set and cleared by software to enable/disable the acquisition (counter is counting) on the corresponding analog I/O group x.

0: Acquisition on analog I/O group x disabled

1: Acquisition on analog I/O group x enabled

19.6.10 TSC I/O group x counter register (TSC_I OGxCR)

x represents the analog I/O group number.

Address offset: 0x30 + 0x04 * x, (x = 1 to 8)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	CNT[13:0]													
		r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:14 Reserved, must be kept at reset value.

Bits 13:0 **CNT[13:0]**: Counter value

These bits represent the number of charge transfer cycles generated on the analog I/O group x to complete its acquisition (voltage across C_S has reached the threshold).

19.6.11 TSC register map

Table 118. TSC register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
0x0000	TSC_CR	CTPH[3:0]			CTPL[3:0]				SSD[6:0]								SSE	SSPSC		PGPSC[2:0]							MCV [2:0]		IODEF	SYNCPOL	AM	START	TSCE				
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					0	0	0	0	0	0	0	0			
0x0004	TSC_IER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCEIE	EOAIE				
	Reset value																															0	0				
0x0008	TSC_ICR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCEIC	EOAIC				
	Reset value																															0	0				
0x000C	TSC_ISR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCEF	EOAF				
	Reset value																															0	0				
0x0010	TSC_IOHCR	G8_IO4	G8_IO3	G8_IO2	G8_IO1	G7_IO4	G7_IO3	G7_IO2	G7_IO1	G6_IO4	G6_IO3	G6_IO2	G6_IO1	G5_IO4	G5_IO3	G5_IO2	G5_IO1	G4_IO4	G4_IO3	G4_IO2	G4_IO1	G3_IO4	G3_IO3	G3_IO2	G3_IO1	G2_IO4	G2_IO3	G2_IO2	G2_IO1	G1_IO4	G1_IO3	G1_IO2	G1_IO1				
	Reset value	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1			
0x0014	Reserved																																				
0x0018	TSC_IOASCR	G8_IO4	G8_IO3	G8_IO2	G8_IO1	G7_IO4	G7_IO3	G7_IO2	G7_IO1	G6_IO4	G6_IO3	G6_IO2	G6_IO1	G5_IO4	G5_IO3	G5_IO2	G5_IO1	G4_IO4	G4_IO3	G4_IO2	G4_IO1	G3_IO4	G3_IO3	G3_IO2	G3_IO1	G2_IO4	G2_IO3	G2_IO2	G2_IO1	G1_IO4	G1_IO3	G1_IO2	G1_IO1				
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x001C	Reserved																																				
0x0020	TSC_IOSCR	G8_IO4	G8_IO3	G8_IO2	G8_IO1	G7_IO4	G7_IO3	G7_IO2	G7_IO1	G6_IO4	G6_IO3	G6_IO2	G6_IO1	G5_IO4	G5_IO3	G5_IO2	G5_IO1	G4_IO4	G4_IO3	G4_IO2	G4_IO1	G3_IO4	G3_IO3	G3_IO2	G3_IO1	G2_IO4	G2_IO3	G2_IO2	G2_IO1	G1_IO4	G1_IO3	G1_IO2	G1_IO1				
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x0024	Reserved																																				
0x0028	TSC_IOCCR	G8_IO4	G8_IO3	G8_IO2	G8_IO1	G7_IO4	G7_IO3	G7_IO2	G7_IO1	G6_IO4	G6_IO3	G6_IO2	G6_IO1	G5_IO4	G5_IO3	G5_IO2	G5_IO1	G4_IO4	G4_IO3	G4_IO2	G4_IO1	G3_IO4	G3_IO3	G3_IO2	G3_IO1	G2_IO4	G2_IO3	G2_IO2	G2_IO1	G1_IO4	G1_IO3	G1_IO2	G1_IO1				
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x002C	Reserved																																				
0x0030	TSC_IOGCSR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.				
	Reset value																																				
0x0034	TSC_IOG1CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[13:0]																
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x0038	TSC_IOG2CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[13:0]																
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0	0	0			

Table 118. TSC register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x003C	TSC_I0G3CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[13:0]													
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0040	TSC_I0G4CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[13:0]													
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0044	TSC_I0G5CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[13:0]													
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0048	TSC_I0G6CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[13:0]													
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x004C	TSC_I0G7CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[13:0]													
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0050	TSC_I0G8CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[13:0]													
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0

Refer to [Section 3.2 on page 53](#) for the register boundary addresses.